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ERGONOMIC MICROBIAL AIR SAMPLE

Abstract

A gas sampler device has a top plate with a concaved outer wall. The concaved outer wall allows users easily to lift the top plate off of the bottom plate without disturbing the bottom plate because the curved surface permits more positive contact between the outer wall and users' fingers. Moreover, the weight of the top plate is reduced by approximately twenty percent compared to conventional top plates, a feature that also makes it easier for users to lift the top plate off of the bottom plate.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a microbiological gas sampler. More particularly, the present invention is for a microbiological gas sampler for use in a controlled environment that includes an ergonomic top plate with concaved sidewalls. The concaved sidewalls allow both more positive contact when attempting to grasp the top plate and a reduction of the weight of the device.

Background of the Related Art

[0002] A controlled environment is an area which is designed, maintained, or controlled to prevent particle and microbiological contamination of products. Controlled environments include, for example, clean rooms and clean hoods. There are different levels of cleanliness in clean rooms, generally in the range of a Class 100 room (i.e., a room having no more than 100 particles of 0.5 micron and larger, per cubic foot of air), to a Class 10,000 clean room.

[0003] Clean rooms are used for a variety of purposes, such as in the manufacture of pharmaceutical products and electronics, such as semiconductors. Often, clean rooms are used to work on extremely expensive and complex products, and it is not unusual that there be millions of dollars worth of product in a clean room at any given time. Clean rooms have to maintain a high level of cleanliness, or risk large financial losses. If a product being developed or manufactured in a clean room becomes contaminated, the entire product in the clean room must often be discarded.

[0004] Microbial air samplers are used to monitor the level of cleanliness (in terms of viable contamination) in a controlled environment. One or more samplers are positioned about the clean room to collect airborne particulates and organisms (or microorganisms) such as bacteria and fungi. Samplers that run at high flow rates permit air to enter the sampler at such high flow rates that loss of smaller particulates carrying microorganisms is normality (i.e., smaller particulates are not retained in the medium). At the same time high flow rate air samplers only sample for a short time period and relay on a short snapshot of the condition of the area. Samplers running at 28.3 LPM (liters per minute) must operate for a longer period of time than a unit running at 322 LPM. In doing this, they sample a broader spectrum of the drug fill time and present superior data as the sample time takes a larger snapshot of the operation. Samplers that run at 28.3 LPM also provide the ability to capture more smaller particulates that may be lost due to dynamic drag (or an umbrella affect) in higher flow rate units.

[0005] Air sampling systems are generally known, and an air sampling system is offered by Veltek Associates, Inc. known as SMA (Sterilizable Microbiological Atrium) Microbial Air Sampler System. One such system is shown in U.S. Pub. No. 2011/0167931, filed Jan. 12, 2010, and U.S. Pat. No. 7,940,188, filed Jul. 26, 2010, the entire contents of which are hereby incorporated by reference. As noted in those applications, the air sampler system includes a controller connected to a vacuum pump to control the flow of air to air sampler devices located in the clean room.

[0006] A prior art air sampler device 5 is shown in FIGS. 1(a), (b), which is offered by Veltek Associates, Inc. The assembled air sampler device 5 includes a top plate 10 with holes 11 and a bottom plate 14. The top plate 10 has a flat section and an outer side. The flat section forms the top surface of the top plate 10 and extends substantially horizontally when in use. The openings pass

through the flat section. The outer side extends downward to be substantially orthogonal to the flat section. The outer side has a single uniform thickness that extends the entire circumference of the top plate **10**. The outer surfaces of the top plate **10** and the bottom plate **14** are flat and smooth. The bottom plate **14** is sized and shaped substantially the same as the top plate **10**. Though the device **5** is shown as circular, other shapes may be used.

[0007] In operation, the top plate **10** is removed, a Petri dish is placed on the bottom plate **14**, and the top plate **10** is replaced on the bottom plate **14**. A vacuum tube is attached to the air port **22**. Air is then sucked in through the holes **11** in the top plate **10**, so that the air strikes a test medium contained in a Petri dish, which is inside the air sampler device **5** between the top plate **10** and the bottom plate **14**. The air exits through the air port **22** and vacuum tube. At the end of the testing period, the top plate **10** is again taken off of the bottom plate **14**, the Petri dish is removed, and the top plate **10** is replaced. The Petri dish can then be analyzed to determine the level of cleanliness of the clean room.

[0008] The entire device **5** is metal so that the device **5** can be sterilized by heat, steam, Vaporized Hydrogen Peroxide (VHP) or Ethylene Oxide (ETO). The Petri dish has a diameter of about 3.5 inches. The top plate **10** has an outer diameter of 4.5 inches. There are twelve holes **11** positioned within about a circular area having a 3 inch diameter, and each hole **11** has a diameter of about 0.5 inches.

[0009] However, the sides of the top plate **10** are smooth and the top plate **10** is relatively heavy, specifically, 1 pound, 4.2 ounces. Consequently, the top plate **10** can be difficult to grasp by a person inside the clean room who is wearing gloves.

SUMMARY OF THE INVENTION

[0010] Accordingly, it is an object of the invention to provide a device for sampling viable cells in air. It is another object of the invention to provide a microbial air sampler having an improved design that both decreases the weight of the top plate and allows for more positive contact when grasping the top plate.

[0011] Accordingly, an air sampler device is provided having a concaved sidewall along the outer circumference of the top plate. The concaved sidewall is particularly useful because users are often required to wear gloves at all times while inside the clean room. The concaved sidewall allows the top plate to be easily lifted off of the bottom plate without disturbing the bottom plate because more positive contact can be made between the fingers of the user and the sidewall of the top plate.

[0012] Moreover, the concaved sidewall decreases the weight of the device by approximately twenty percent. The decreased weight of the top plate also makes it easier for users to lift the top plate off of the bottom plate.

[0013] These and other objects of the invention, as well as many of the intended advantages thereof, will become more readily apparent when reference is made to the following description, taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0014] FIG. **1(a)** is a perspective view of the air sampler device in accordance with the prior art;

[0015] FIG. **1(b)** is a cross section view of the air sampler device in accordance with the prior art;

[0016] FIG. **2** is a top perspective view of the top plate of the air sampler device in accordance with an exemplary embodiment of the invention; and

[0017] FIG. **3** is a cross-section side view of the air sampler device of FIG. **2** showing movement of air within the device.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0018] In describing a preferred embodiment of the invention illustrated in the drawings, specific

terminology will be resorted to for the sake of clarity. However, the invention is not intended to be limited to the specific terms so selected, and it is to be understood that each specific term includes all technical equivalents that operate in similar manner to accomplish a similar purpose. Preferred embodiments of the invention are described for illustrative purposes, it being understood that the invention may be embodied in other forms not specifically shown in the drawings.

[0019] As shown in FIG. 2, a top plate **100** of an air sampler device **50** is shown. The top plate **100** has a top surface **112**. There are twelve holes **116** formed in the top surface **112**. The top plate **100** has a flat section **130** and at least one outer side **140**. The flat section **130** forms the top surface of the top plate **100** and extends substantially horizontally when in use. The openings **116** pass through the flat section **130**. The outer side or side wall **140** extends downward to be substantially orthogonal to the flat section **130**. The side wall **140** is concaved or curved inward and defines the entire outer circumference of the top plate **100**. The concaved outer side **140** makes the top plate **100** easy to grip, so that a user can easily remove and replace the top plate **100** with respect to the bottom plate **150**.

[0020] For illustrative purposes, the bottom plate **150** is also shown in FIG. 3. A vacuum air port **160** is positioned at the side of the bottom plate and communicates with an air hole **162**. The air hole **162** extends through the bottom plate **150**, from the air port **160** to the center well **158**. The vacuum air port **160** connects to a vacuum tube to draw air through the sampler **50**.

[0021] The operation of the sampler **50** is best shown in FIG. 3, where the arrows generally show the direction of travel of the air as it flows through the device **50**. A sterilized air sampler device **50** is introduced into the clean room, and the top plate **100** is removed. The Petri dish **52** is inserted onto the bottom plate **150**, and the top plate **100** is replaced. The air flow is then initiated for a predetermined period of time. Air is drawn into the sampler device **50** by the vacuum tube **162** through the air port **160**.

[0022] Once the air enters the holes **112** in the top plate, it strikes the capture material in the Petri dish **52**, then travels up around the sides of the Petri dish **52**, through the elongated slots **164** beneath the Petri dish **52**, and enters the center well **158**. The air is then sucked through the air hole **162** and exits out of the vacuum air port **160**. Once the predetermined period of time (which can be from 10-60 minutes or longer) has lapsed, the air flow is turned off. The top plate **100** is then raised, and the Petri dish **52** is removed for testing. The sampler **50** can then be sterilized, if desired, and a new Petri dish **52** introduced.

[0023] Accordingly, the air port **160** is in flow communication with the passageway **162**, which is in flow communication with the well **158**. And, the well **158** is in flow communication with the elongated slots **164**, which are in flow communication with air entering the holes **112** in the top plate **100**. The structure and operation of the device having the bottom plate **150** shown, is more fully described in Pub. No. 2011/0167931, which is incorporated herein by reference. It should be noted, however, that any suitable bottom plate **150** can be provided other than the one shown, and the bottom plate **150** need not have a center well **158** and air hole **162** and slots **164**.

[0024] As shown in FIG. 3, the side wall **140** defines a side wall inner surface **170** and a side wall outer surface **172**. The inner surface **170** defines an inner diameter of the upper plate **140**. The inner surface **170** is straight, and does not project inward, so as not to interfere with the Petri dish **52**. Thus, the inner wall **170** need not be positioned further away from the Petri dish **52**. The straight inner wall **170** also provides a straight air conduit with uniform dimensions (i.e., width) between the Petri dish **52** and the inner wall **170**, so that air can flow uniformly around the Petri dish **52** to the exit port **160**.

[0025] The outer surface **172** of the side wall **140** is curved, and therefore the side wall **140** has a varying thickness. The side wall **140** is thicker at the top and bottom portions, and thinner in the middle portion where it is curved inward. The outer surface **172** is curved to be ergonomic and mate with the shape of a user's gloved hand. The top of the curved outer surface **172** has a curved top lip **142** that extends around the entire circumference of the top plate **100**. The top lip **142**

projects outward from the middle portion of the wall **140**. The top lip **142** engages the user's fingers when the top plate **100** is being lifted, thereby making it easier to lift and manipulate the top plate **100**. The top lip **142** is rounded or beveled at the end **143** between the top surface of the flat section **130** and the side wall **140**. The rounded end **143** enhances the safety of the top plate **100** and increases the comfort of the user, by removing any sharp angles. An additional benefit of the curved outer wall surface **172** is that the user knows by touch that he is manipulating the top plate **100** (and not the bottom plate **150**). Thus, for instance, the user can slide his/her fingers up from the bottom of the device **50** until the fingers engage the top lip **142**.

[0026] The bottom section of the curved outer wall surface **172** forms a bottom lip **144** that is relatively sharp. This provides the side wall **140** with a flat foot **146** that forms a seal and mates with the top surface of the bottom plate **150**. The outer circumference of the foot **146** of the side wall **140** where the top plate **100** meets the bottom plate **150** is sized and configured substantially the same as the bottom plate **150**. The foot of the side wall **140** of the top plate **100** has an outer diameter of 4.5 inches. There are twelve holes **116** positioned within about a circular area having a 3 inch diameter, and each hole **116** has a diameter of about 0.5 inches. The thickness of the top and bottom of the concaved side **140** is about 0.25 inches.

[0027] The concaved outer side **140** of the top plate **100** is particularly useful since users are often required to wear gloves (in addition to garments, hoods, and booties) at all times while inside the clean room. In addition, the entire device **50** is made of metal, so that it can be sterilized by heat, steam, VHP or ETO. Consequently, the top plate **100** (as well as the bottom plate) is relatively heavy, which makes it difficult to remove with a gloved hand. By providing an inwardly concaved outer side **140**, the present invention allows the top plate **100** to be easily lifted off of the bottom plate **150** without disturbing the bottom plate **150**.

[0028] The curved side wall **140** also substantially reduces the weight of the top plate **100**. The weight of the top plate **100** is 1 pound, 1 ounce, which is about a 20% (18.82%) reduction in weight from the top plate **10** of FIG. 1, which weighs 1 pound, 4.2 ounces. The radius of curvature is about 0.337 inches.

[0029] The curved outer surface **172** is preferably uniformly curved completely around the top plate **100**. However, the outer surface **172** need not be completely curved and only portions of the outer surface **172** can be curved. For instance, the side wall **140** can be curved at two opposite portions of the top plate **140**. Or, the side wall **140** can be curved differently at different portions of the side wall **140**. And, the curve need not have both a top lip **142** and a bottom lip **144**. For instance, only a top lip **142** can be provided.

[0030] The foregoing description and drawings should be considered as illustrative only of the principles of the invention. The invention may be configured in a variety of shapes and sizes and is not intended to be limited by the preferred embodiment. Numerous applications of the invention will readily occur to those skilled in the art. Therefore, it is not desired to limit the invention to the specific examples disclosed or the exact construction and operation shown and described. Rather, all suitable modifications and equivalents may be resorted to, falling within the scope of the invention.

Claims

1. A gas sampler device comprising: a top plate having a top section and a downward turned side with an inner surface and an outer surface, wherein the inner surface is straight and substantially orthogonal to the top section and the outer surface is concaved inward; and, a bottom plate having a top surface and a receiving portion at the top surface for receiving a dish, the top surface of said bottom plate mating with the side wall of said top plate.
2. The gas sampler device of claim 1, the side having a top portion with a beveled outer corner formed between the side and the top section.

3. The gas sampler device of claim 1, the side having a top lip that is configured to engage a user's finger.
 4. The gas sampler device of claim 3, the side having a bottom lip.
 5. A top plate for a gas sampler device, said top plate having a concaved outer wall that curves inward around the outer circumference of the top plate.
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